Four Rational Parking Functions, Not Twelve
Counterexample to Conjecture 00000000456

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AI-assisted proof and reproduction; no affiliation claimed

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Abstract

The stated factorial ratio is not the cardinality of standard rational parking functions. In the height-first convention, parameters $(m, n) = (3, 2)$ give four labelled rational Dyck paths, while the displayed ratio is $4!/2! = 12$. An enumerative positive q-refinement cannot specialize at $q = 1$ both to the actual count and to this ratio. A self-contained pinned Lean project verifies actual paths, every-vertex diagonal tests, exhaustive distinct labels, the path/column bridge, the four-object count and the closed specialization negation.

1 The named objects and the necessary clause

Both source languages name rational parking functions and describe the factorial ratio as their count. The Chinese explicitly states the counting formula; the English calls the requested positivity a strengthening of that formula. Following the height-first convention of [1], an (m, n) rational parking function is a Dyck path from $(0, 0)$ to (n, m), with its m north steps labelled bijectively by $1, \ldots, m$, increasing from bottom to top within each column. We use positive coprime parameters $(m, n) = (3, 2)$.

A genuine enumerative q-analogue Q must satisfy $Q(1) = |\mathrm{PF}_{m,n}|$. The stated counting formula would simultaneously require

$$Q(1) = \frac{(m + n - 1)!}{(m - 1)!}.$$

This condition is independent of the chosen statistic. As a bare number, the factorial ratio does have a positive q-integer product; that numerical fact is not denied here. Such a product is not a refinement of the actual object count in the counterexample. Dropping the object count would change the assertion.

2 An exhaustive geometric count

Let c_i be the column where the ith north step starts, for $i = 0, 1, 2$. The path lies above the line $3x = 2y$ precisely when

$$0 \le c_0 \le c_1 \le c_2 < 2, \qquad 3c_i \le 2i.$$

Thus (c_0, c_1, c_2) is either $(0, 0, 0)$ or $(0, 0, 1)$. The column data reconstruct the unique word by inserting east runs before, between and after the three north steps. Each horizontal run has its minimum diagonal margin at the right endpoint; north steps increase the margin. The terminal east run at height three ends at width two and stays above the line. This accounts for every vertex and the endpoint, not just north steps.

For $(0, 0, 0)$ all three labels must increase, giving one labelling. For $(0, 0, 1)$ only the first two must increase, giving three labellings:

word	north labels in path order	preferences in label order
$NNNEE$	$(1, 2, 3)$	$(0, 0, 0)$
$NNENE$	$(1, 2, 3)$	$(0, 0, 1)$
$NNENE$	$(1, 3, 2)$	$(0, 1, 0)$
$NNENE$	$(2, 3, 1)$	$(1, 0, 0)$

Preferences record the north-step column for each label. Sorting columns and sorting labels within each column reconstruct the labelled path; hence no object or multiplicity is hidden by this representation.

Theorem 1. *For standard rational parking functions at height-first parameters* $(3, 2)$*, the actual count is four and the stated factorial ratio is twelve. Consequently the claimed enumerative positive* q*-refinement cannot exist.*

Proof. The exhaustive count above gives four distinct objects, whereas $(3+2-1)!/(3-1)! = 24/2 = 12$. If a required refinement existed, its value at one would be both four and twelve, a contradiction. □

The convention that lists width first also cannot rescue a universal formula: the same paths at $(2, 3)$ would give four against twenty-four. The Lean proof advertises only the declared height-first witness, not an unverified extra case.

3 Kernel-checked definition bridges

The Lean source imports only the `Std` library bundled with Lean 4.33.1. Actual north/east words have a recursive vertex test and exact step counts. The theorem `word_universe` covers every length-five word; `raw_word_bridge` and `path_injective` identify the path and column representations without omitting terminal steps. Every raw column/label object is enumerated, membership is equivalent to the actual validity predicate, the four-element list is exact and has no duplicates. Distinct labels in the three-element type use all labels; the same-column strict inequalities are enforced before counting.

Finite natural-coefficient lists represent positive polynomials, with ordinary Horner evaluation. The theorem `eval_at_one` identifies evaluation with the coefficient sum. The final closed specialization theorem rules out simultaneous evaluation to the genuine count and the proposed formula. Kernel `decide` is used only on actual finite words, labels and their fully specified predicates, not as a replacement for a definition bridge. Every advertised theorem has its actual type and transitive axioms printed.

4 Reproducibility and limits

The accompanying Python reproducer independently checks all endpoint-compatible words and label permutations, then cleanly rebuilds this pinned project and audits every theorem. It has no external package dependencies or network bootstrap requirement. Compiler binaries and bundled standard-library modules remain trusted; no compiler audit, cross-platform run or human expert endorsement is claimed. The submitted PDF is actually compiled from this source; validation also recompiles and compares its bytes.

This is a counterexample to a necessary counting clause, not a denial of other parking-function positivity theorems or of the bare numerical q-integer product. No source conjecture or organizer metadata is modified, and no affiliation, first-solution credit, score or upstream acceptance is asserted.

References

[1] D. Armstrong, N. A. Loehr, G. S. Warrington, *Rational Parking Functions and Catalan Numbers*, arXiv:1403.1845v1 (2014), Sections 3.1–3.2 and Corollary 4.